

Regular and Supplementary Winter Examination – 2024

Branch: Civil Engineering

Course: B. Tech.

Semester : V

Subject Code & Name: BTCVC503 Structural Mechanics – II

Max Marks: 60

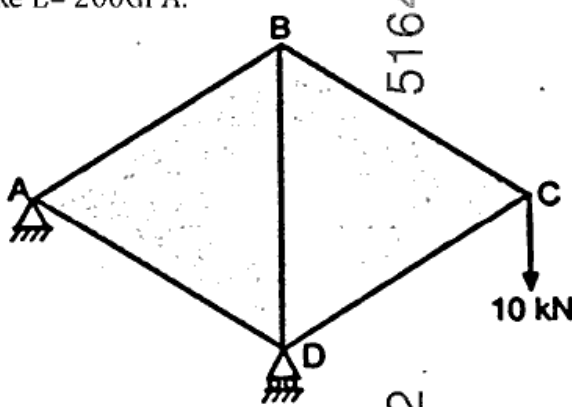
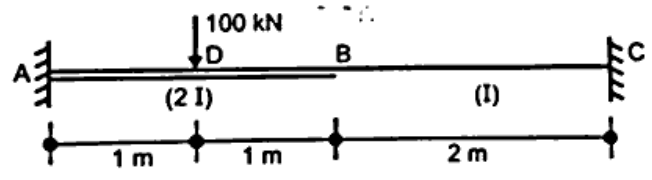
Date:11/02/2025

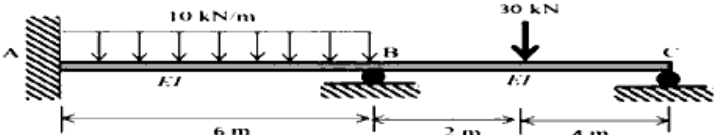
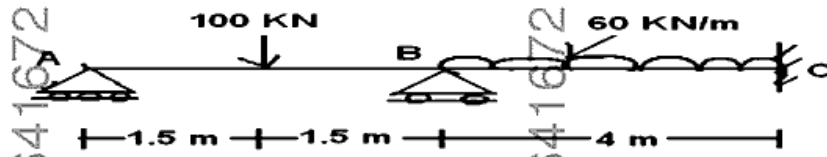
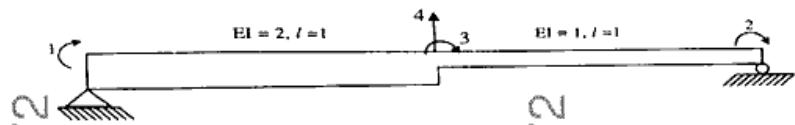
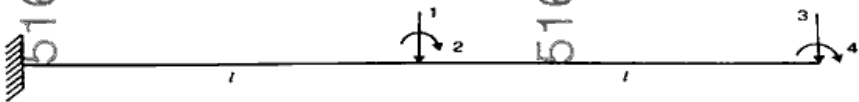
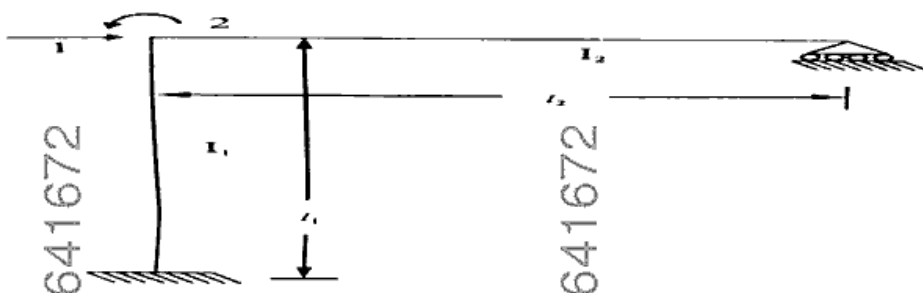
Duration: 3 Hr.

Instructions to the Students:

- Each question carries 12 marks.
- Question No. 1 will be compulsory and include objective-type questions.
- Candidates are required to attempt any four questions from Question No. 2 to Question No. 10.
- The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
- Use of non-programmable scientific calculators is allowed.
- Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q.1	Objective type questions. (Compulsory Question)		12
1	If there are 'm' unknown member forces, 'r' unknown reaction components and 'j' number of joints, then degree of static indeterminacy of a pin jointed plane frame is given by a. $m + r + 2j$ b. $m - r + 2j$ c. $m + r - 2j$ d. $m + r - 3j$	1	1
2	If there are 'm' unknown member forces, 'r' unknown reaction components and 'j' number of joints, then a pin jointed plane frame is unstable if a. $(m + r) > 2j$ b. $(m + r) = 2j$ c. $(m + r) < 2j$ d. None of the above		1
3	For drawing ILD, what value of test load is assumed? a. 1 Unit b. Arbitrary c. Depends upon structure d. 0		1
4	ILD of statically determinate beams consists of curve as well as straight lines. a. True b. False	1	1
5	Cable is a tension member a. True b. False	2	1
6	The shape of the cable, when loaded with uniformly distributed load throughout the span is a. Linear always b. Parabolic always c. Trapezoidal always d. Linear or parabolic		1
7	Which of the following conditions form the basis of Direct Flexibility Method a. Equilibrium Conditions b. Force-Displacement conditions c. Load Applied d. Compatibility Conditions		1
8	Flexibility coefficients are used in which of the following method? a. Force method b. Displacement method c. Both force and displacement method d. Virtual force method	3	1
9	Stiffness coefficients are used in which of the following method? a. Force method b. Displacement method c. Both force and displacement method d. Virtual force method	4	1
10	Which of the following conditions form the basis of Direct Stiffness Method a. Equilibrium b. Force- c. Load Applied d. Compatibility	4	1

	Conditions	Displacement conditions		Conditions		
11	If the structure is divided into discrete areas or volumes then it is called as				5	1
	a. Structure	b. Element	c. Matrix	d. Boundaries		
12	One dimensional element is the linear segments which are used to model				5	1
	a. Bars and trusses	b. Plates and beams	c. Structures	d. Solids		
Q. 2	Solve the following.					12
A)	A truss consisting of two equivalent triangles is loaded as shown in figure. Determine the horizontal deflection of roller at D. The length of each member is 3m. The C.S. areas of all ties are 300mm ² each and that of all struts are 600mm ² each. Take E= 200GPa.				1	6
						
B)	A simply supported beam CD has span 12m. It is loaded with UDL 45kN/m, span 5m, and starts 3.5m from C. A concentrated load of 50kN is acting at 2m from D. Find Shear Force at 5m from C by ILD method.					6
Q.3	Solve the following.					12
A)	A light cable is supported at two points 20m apart, which are at same level. The cable supports three equidistance point loads of 40kN, 30kN and 20kN from left support. The deflection at first point is found to be 0.8m. Determine the tension in the different segments and total length of Cable.				2	6
B)	A three hinged parabolic arch of 20m span and 3m rise is carrying a point load of 100kN at a section 5m from the left support. Find the value of horizontal thrust and bending moment at a point 7.5m from the right support.					
Q. 4	Solve Any Two of the following.					12
A)	Analyze the beam shown in figure by flexibility method.				3	6
						

B)	Analyze the beam shown in figure by flexibility method if support B sinks by 25mm. Take $EI=3800\text{KNm}^2$	3	6
			
C)	Analyse the beam shown in figure by flexibility method.	51641672	6
			
Q.5	Solve Any Two of the following.		12
A)	Develop the stiffness matrix of the beam shown with respect to the 04 degrees of freedom given.	4	6
			
B)	Develop the stiffness matrix of the beam shown with respect to the 04 degrees of freedom given.	51641672	6
			
C)	Develop the stiffness matrix of the frame shown with respect to the given coordinates.	4	6
			
Q. 6	Solve Any Two of the following.		12
A)	Write short note on types elements used in finite elements analysis method.	5	6
B)	Write short note on Plane Strain and Axi- symmetric Problem in finite elements analysis method.	5	6
C)	Write short note on shape functions used in finite elements analysis method.	5	6
*** End ***			